Insurance Claim Denial Analysis Jan 2025-May 2025

Revenue Cycle Insurance Denial Trend and Driver Analysis

Excel, Power Query, SQL Server, and Power BI

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Business Question & Why It Matters

Where are claim denials coming from?

- What are the top drivers by payer and reason?
- How are denials trending over time?

Why does this matter?

- Denials delay reimbursement
- Denials create rework and operational waste

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Data and Scope

Data Source

- Kaggle: **CA Hospital Dataset – Q1 2025** [1]

Scope

- Insurance claims only
- One row per claim

Limitations

- January and May contain partial-month data
- Synthetic data

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Workflow

Excel/Power Query

- Data exploration
- Data validation
- Data cleaning
- Dashboard prototyping

SQL Server

- Built reusable claim-level base view
- Validated row counts and denied claims

Power BI

- Created measures
- Built final dashboard

Documentation

- README
- Definitions
- Change Log

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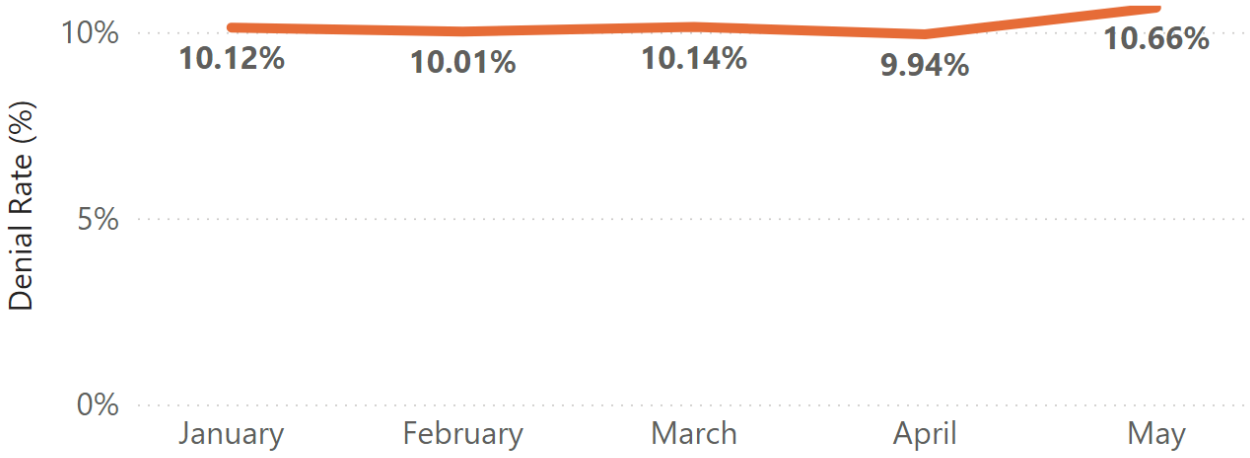
Insurance Claim Denial Analysis
Jan 2025-May 2025

Monthly Denial Rate Trend



Note: January and May contain partial month data.

Overall Denial Rate

10.06%

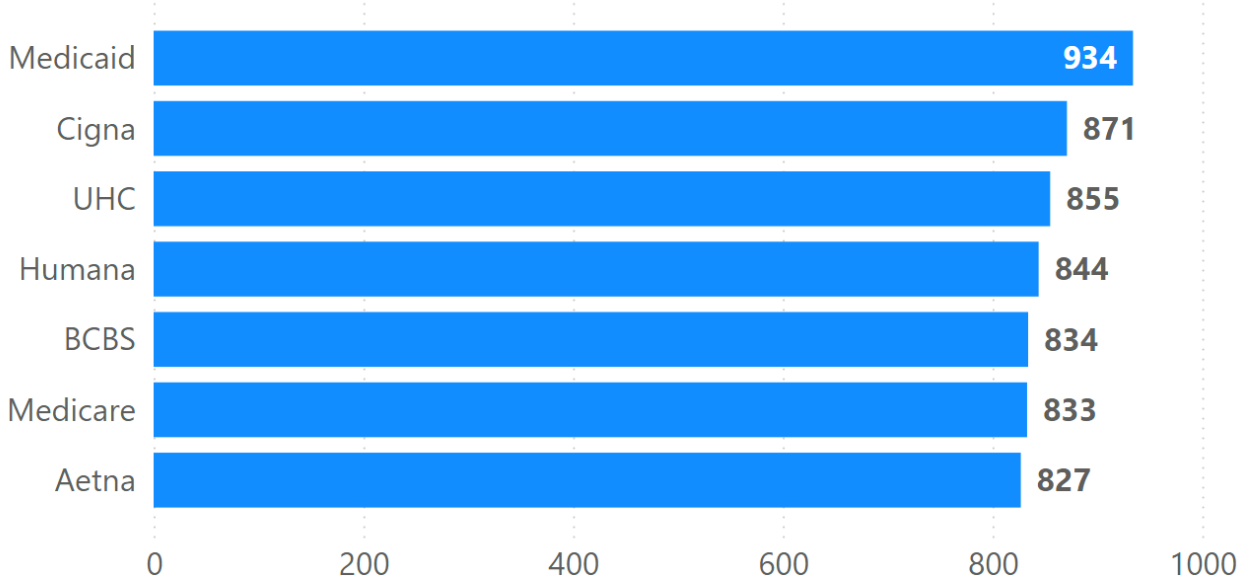
Top Denial Reason

Duplicate Claim

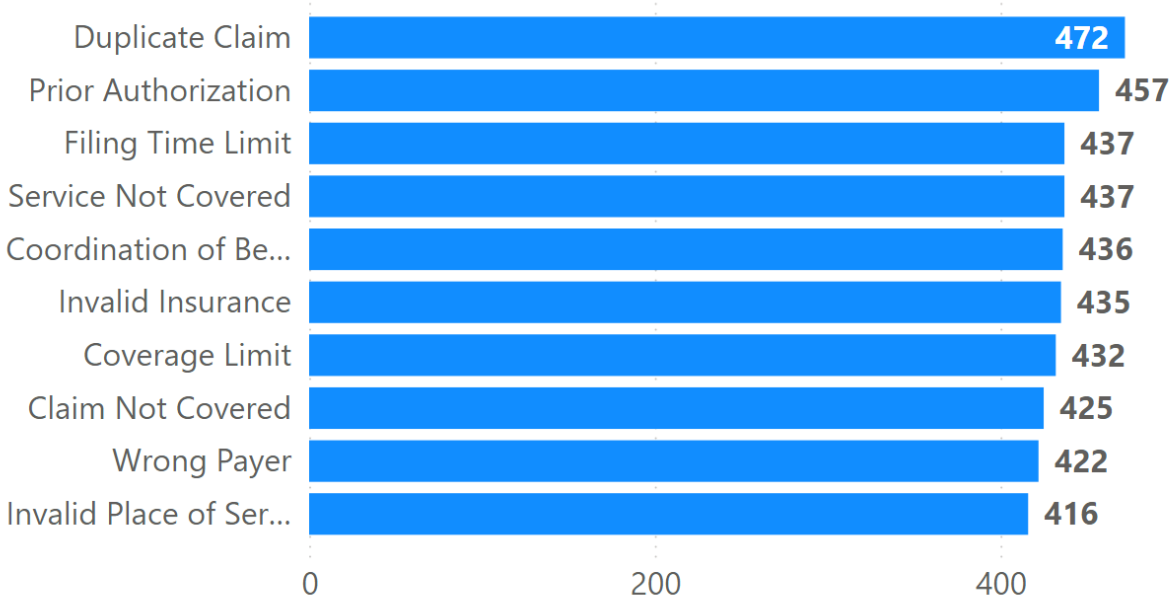
Top Payer by Denied Volume

Medicaid

Denied Claims by Payer



Top 10 Denial Reasons by Denied Claims



Key Findings & Next Steps

Top 3 Denial Reasons

- Duplicate Claim
- Prior Authorization Required
- Timely Filing Limit Exceeded

Process Improvement Opportunities

- Investigate root causes of duplicate claims
- Strengthen prior authorization checks before claim submission
- Improve timely filing compliance

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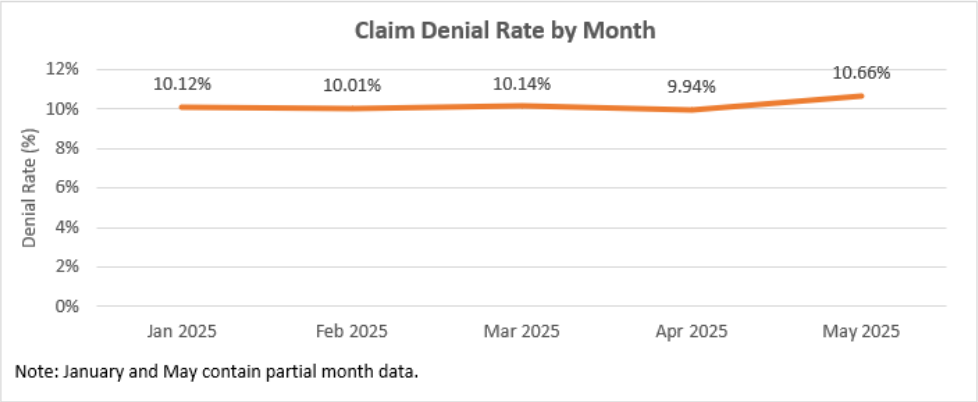
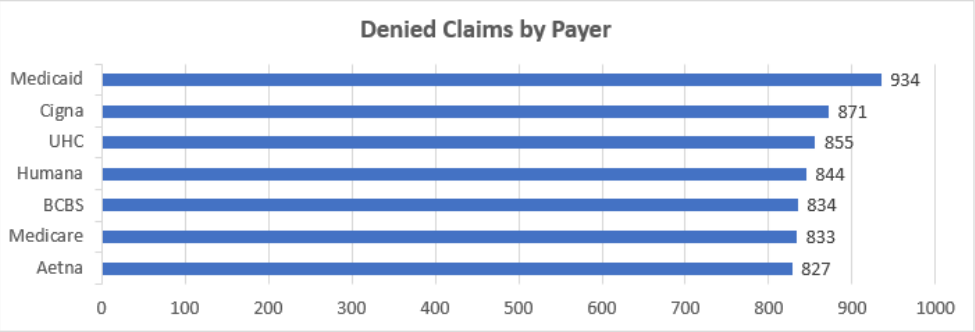
Extending the Healing Ministry of Christ

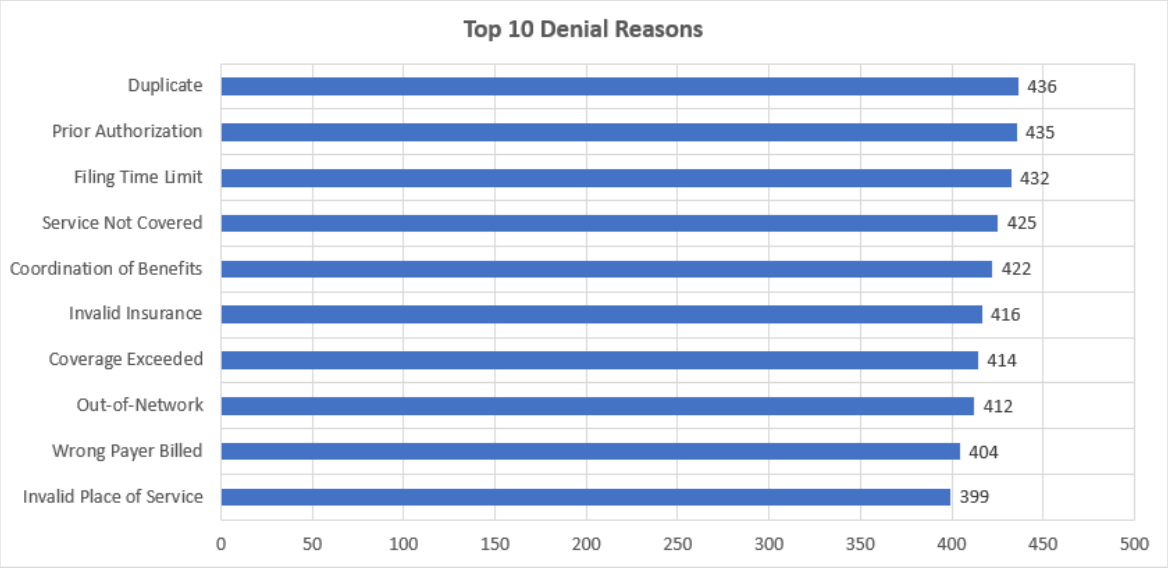
Appendix A: Excel Dashboard

Insurance Claim Denial Analysis
Jan 2025 - May 2025





Overall Denial Rate:	Top Payer by Denied Volume:	Top Denial Reason:
10.06%	**Medicaid**	**Duplicate Claim**



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Appendix B: References

[1] Kaggle Dataset: CA Hospital Dataset – Q1 2025

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